

MOAZ HANY ELBAKHSHA

Electronics Mechatronics Engineer

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PROFESSIONAL SUMMARY

Final-year Mechatronics Engineering student specializing in closed-loop control systems, precision electromechanical interfaces, and high-reliability multi-layer PCB design. Industry experience at ViE Technologies developing encoder-driven automated moving bases and closed-loop illumination test benches, alongside engineering 4-layer STM32H7 central mainboards and 14-bit magnetic encoder boards for international robotics competitions. Dedicated to delivering robust electromechanical control systems, test automation, and embedded solutions.

CORE SKILLS

Control Systems & Automation: PID Control, Motion Control, Motor Controllers/VFDs, Closed-Loop Positioning, Encoder Feedback, Sensor Arrays, MATLAB/Simulink

Hardware & PCB Design: Altium Designer, KiCad, 4-Layer PCB Design, System Schematics, Power Electronics (50A/24V, TPS eFuses, LDOs), Signal Integrity, DFM

Embedded Firmware & Fieldbus: C/C++, Python (Test Automation), STM32 (H7/F4/F1), Arduino, Microcontrollers, Isolated CAN Bus, SPI, I2C, UART, Ethernet, Linux/Bash, ROS2

CAD & Schematics: AutoCAD (Electrical Schematics, Control Networks, Panel & Enclosure Layouts), SolidWorks 3D Modeling, Prototype Fabrication

Tools & Methodologies: Logic Analyzer, Multimeter, Git, Cross-Functional Engineering Collaboration, Design Reviews, Equipment Fitting

EXPERIENCE

Embedded Systems & Python Engineering Intern — Camera Team

Jul 2025 – Oct 2025

ViE Technologies

Penang, Malaysia

- **Precision Electromechanical Control:** Designed and programmed an encoder-driven, limit-switch-protected motorized moving base for automated camera QA test rigs, significantly boosting positioning repeatability and motion stability.
- **Closed-Loop Dynamic Illumination:** Engineered a closed-loop photodiode-based illumination control system with non-linear intensity compensation, communicating sensor feedback to programmable power supplies and eliminating manual line calibration.
- **Automated Test Bench Integration:** Built an automated optical QA test bench with custom Python control software for X-ray protective glass inspection, handling hardware selection, enclosure fitting, and schematic integration.

PCB Designer & Electronics Lead Member

Oct 2023 – Oct 2025

UTM Robocon Club — Universiti Teknologi Malaysia

Johor, Malaysia

- **4-Layer Central Robotics Mainboard:** Engineered a high-density 4-layer STM32H7 mainboard in Altium Designer featuring 3x isolated CAN buses, 3x UART, SPI, dual I2C, 6x PWM motor outputs, dual QEI encoder interfaces, and onboard TPS eFuse power protection.
- **Precision Encoder & Localization PCBs:** Designed a 14-bit magnetic rotary encoder board (AS5047P) with an onboard LDO to step down the robot's 12V bus directly to 3.3V, alongside a 2-layer localization coprocessor board offloading odometry telemetry over dual CAN bus to central controllers.
- **Industrial Power Control & Troubleshooting:** Engineered a 5V signal-controlled 50A/24V motor power relay cutoff module (AZ21501) with optocoupler isolation, and diagnosed critical communication/sensor faults under competition time pressure across 2 international events.

KEY PROJECTS

Smart Automation System Using Robotic Arm & AI Vision — *Mechatronics Theme Award*

Oct 2025 – Feb 2026

- Developed an intelligent pick-and-place workcell integrating an industrial robotic arm, YOLO object detection (5 classes), and servo-actuated sorting mechanisms; implemented safety automation with sub-second motor cutoff upon human intrusion.

Autonomous Navigation & Control System for Mobile Robot — Final Year Project

2025 – 2026

- Architected a distributed control framework (PC planning + microcontroller motor execution) fusing ultrasonic arrays, wheel encoders, and MPU6050 IMU data via PID control, and programmed a ROS state machine executing 90° row turns and obstacle recovery.

HONORS & AWARDS

- **Mechatronics Theme Award** — Smart Automation System Using Robotic Arm & AI Vision (UTM 2026)
 - **Panasonic Award** — ABU Robocon 2024 International Robotics Competition (Vietnam)
 - **Champion** — Malaysia Robocon Friendly Match 2024 | **Participant & Competitor** — UTM RBC Autocar Challenge

EDUCATION

Bachelor of Mechatronic Engineering

Oct 2022 – Feb 2027 (Expected)

Universiti Teknologi Malaysia (UTM), Johor, Malaysia